

ROM — Operation Manual

Your offline emergency knowledge appliance

This manual is written for anyone — no technical background needed. Keep a **printed copy with the device**. If the screen and network work but you forget everything else, the two things to remember are: **(1) power it on, (2) join its Wi-Fi and open the page**.

1. What this device is

ROM is a small computer that holds a huge offline library — medical guides, survival and repair manuals, maps, radio instructions, and an encyclopedia — and an assistant named **Rom** that answers your questions and shows you which document the answer came from.

It needs no internet. It makes its own Wi-Fi network. Any phone, tablet, or laptop can connect and use it.

2. Turn it on

1. Connect power (wall outlet, power bank, or solar station).
2. Press the power button. Wait about **1–2 minutes** for it to start.
3. A steady light means it's ready. (On the mini-PC/laptop, the screen will show a login or

a ready message — you do **not** need to log in to use the library from another device.)

3. Connect a phone or laptop to it

1. On your phone, open **Wi-Fi settings**.
2. Join the network named **ROM-LIBRARY**.
3. Password: the one you set (default `emergency-library` — **change it**, see §8).
4. Open any web browser and go to:

> **`http://10.42.0.1:8080`**

(You can also try **`http://rom.library:8080`**.)

1. You'll see the **Rom** chat screen. That's it — you're in.

To browse the raw library instead of asking Rom, go to **`http://10.42.0.1:8090`**.

4. Ask Rom a question

- Type a normal question, like *"How do I treat a burn?"* or *"How do I make water safe to drink?"*, and press **Ask**.

- Rom answers **from your library** and lists its **sources** underneath. Tap a source to open the full original document.

- Green-tagged sources are **clinical / evidence-based**; orange-tagged sources are **traditional / holistic** (not clinically proven). Rom keeps them separate on purpose.

If Rom doesn't know, it will say so rather than make something up. That's by design.

5. Health & safety — read this

- Rom is a **reference tool, not a doctor**. In a serious situation it will tell you to seek a professional or emergency responder — **do that whenever possible**.

- Watch for the " **Get professional help if...**" notes Rom adds to medical answers. Those list the danger signs that mean *stop self-treating and get real help*.

- Never take a medication dose from memory or guesswork. If Rom cites a dose, ****open the cited source and confirm it.****

- Traditional/holistic remedies are included for completeness. They are **not** a substitute for evidence-based care when care is available.

6. Power & battery

- The device uses roughly **5–90 watts** depending on your model (a Raspberry Pi sips power; a gaming-laptop build with the GPU busy uses the most, especially while Rom is "thinking").

- On a large USB-C power bank or power station it will run for **hours**; on wall power or solar, indefinitely.

- To save power: use the **library** (§3, port 8090) instead of asking Rom — browsing uses far less energy than the AI.

- **Shut down cleanly** when you can (don't just yank power): on the mini-PC/laptop choose

Shut Down; on a Pi, hold the power button or run `sudo shutdown now`. This protects the storage.

7. Updating or adding content (needs internet, one time)

Do this **while you still have internet**, before you need the device off-grid.

1. Add more libraries: `./scripts/02-download-content.sh` and pick a pack.
2. Add your own PDFs (guides, manuals, **your personal/legal documents**): copy them into the

`content/` folder.

1. Re-index so Rom can find the new material:

```
docker compose exec rom python ingest.py
```

Keep a **second copy** of the `content/` folder on another drive. Losing the storage is the most likely way to lose everything — a backup fixes that.

8. Change the Wi-Fi password (do this once)

The default password is weak on purpose so you can get started. To set your own:

```
sudo ROM_WIFI_PASS='your-strong-password' ./scripts/01-setup-wifi-hotspot.sh
```

9. Quick troubleshooting

Problem	Try this
Can't find the <code>ROM-LIBRARY</code> Wi-Fi	Wait 2 min after power-on; make sure the device is on; re-run the hotspot script.
Joined Wi-Fi but the page won't load	Confirm the address http://10.42.0.1:8080 (http, not https). Turn phone mobile-data off.
Rom says the library isn't indexed	Run <code>docker compose exec rom python ingest.py</code> .
Rom is very slow	Normal on small devices. Use a smaller model (<code>ROM_MODEL</code> in <code>.env</code>) or just browse the library at <code>:8090</code> .
Rom gives a weird/blank answer	Make sure the model was pulled: <code>docker compose exec ollama ollama list</code> . Re-run <code>./scripts/03-start.sh</code> .

Problem	Try this
Check overall health	Open http://10.42.0.1:8080/health — it lists what's working.

10. One-page emergency cheat sheet

1. **Power on.** Wait ~2 minutes.
2. Phone → **Wi-Fi** → join **ROM-LIBRARY** .
3. Browser → **<http://10.42.0.1:8080>**
4. **Ask Rom** your question. Check the **sources**. Heed the  **warnings**.
5. For less battery use, browse the library at **<http://10.42.0.1:8090>**.
6. Serious injury/illness and help is reachable? **Get professional help.**

Rom is a reference tool, not a substitute for a trained professional. Verify anything safety-critical against the cited source.